Chapter 5
Knowledge Management in Emergent Amateur Organizational Cultures:
Observations From Formula SAE Student Engineering Teams

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ABSTRACT

This chapter examines issues of knowledge management and cultural knowledge in the context of Formula SAE student engineering teams. Approximately 500 student teams field a small formula-style racecar in a series of annual competitions held globally. Despite being small, student-run teams with limited resources and high organizational turnover, strong teams have developed strategies to sustain knowledge creation and work to build the team's cultural knowledge over multiple annual design cycles. This chapter highlights three knowledge management challenges: organizational renewal due to graduation of senior members, capturing vital yet departing tacit and explicit knowledge, and engaging multi-year and collaborative projects. The chapter recommends that strong faculty and institutional support can help FSAE teams develop into stable knowing organizations with deep tacit, explicit, and cultural knowledge bases.

INTRODUCTION

This chapter looks at how organizational culture and cultural knowledge influence knowledge creation and decision making in a specific and somewhat unique domain – student engineering racing teams. These teams organize to design, manufacture, test, refine and field a fully-functional small formula-style autocross racecar in intercollegiate competition, conforming to the standards set by the Society for Automotive Engineers (SAE, 2021b). Such teams and their competition will be henceforth referred to as FSAE for short.

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Compared to larger, more diversified organizations, these student-led organizations are smaller and coping with limited human and financial resources. They are also consistently reinventing themselves due to high turnover due to student graduation. Because of their student membership, participants are also in the process of learning their professional craft as well as how to manage a complex, year-long systems engineering project. These factors lead to particular challenges that can frustrate the team's hopes for success in competition.

This unique contextual environment poses particular challenges to knowledge management and creating a sustained organizational culture. This chapter examines three particular issues teams face and shares observations from the field based on interviews and surveys conducted with student team members and faculty advisors. This chapter concludes that increased institutional support and more active faculty advisors can provide teams with a strong footing to develop cultural knowledge and give them a competitive advantage over schools with newer, more fragmented and more ad hoc organizational structures.

BACKGROUND

Schein defines organizational culture as a pattern of shared basic assumptions that assists the group with adaptation and integration, and are perceived by the group to be worthy of dissemination to new members as the correct way to perceive the group's problems and challenges (Schein, 1992). Organizational culture is the milieu which defines how an organization perceives and addresses problems and affects the overall working climate and behavior of the organization (Saffold, 1988).

With respect to knowledge management, organizational culture defines the social context of what knowledge has value, how new knowledge should be created, validated and shared, and what tools and technologies are best suited to capture and disseminate knowledge created by the organization (De Long & Fahey, 2000). While knowledge management does investigate the use information technology as a facilitative tool to capture and disseminate explicit knowledge, knowledge management research has evolved to understand the overall information ecology of an organization (Nonaka & Konno, 2008). An organization's cultural knowledge helps define a collective identity, purpose and orientation of an organization and informs sensemaking, knowledge creation and decision making, the core information challenges of a knowing organization (Choo, 2006).

While organizational culture is not immutable, attempts to change information discovery, creation, and use without consideration of existing cultural practices risks creating significant crisis points which can derail even the most necessary of change goals (Hayes, 2018). A solid understanding of an organization's cultural knowledge and organizational culture helps shape knowledge management efforts that best fit existing practice and mitigates the consequences of poor fit.

Organizational culture is complex and can vary among and within specific organizations. Martin (1992) notes three perspectives of organizational culture – integration, differentiation and fragmentation. In an integrated organizational culture, there is a high degree of consensus, clarity and consistency, but at the potential expense of groupthink and rigidity. For example, an integrated organizational culture may find internal or external change threatening and respond by doubling down on core values rather than adapting to new circumstances. At the opposite pole, a fragmented organizational culture revels in complexity, with consensus, clarity and consistency being tenuous, fluid and tied to immediate and pressing concerns. While more adaptable to change, such organizations risk being too conflicted to en-

gage change coherently, and may find change sabotaged from within by subcultures holding discordant values or priorities.

Differentiated organizational cultures occupy a middle space where cultural consensus is not absolute and in which multiple subcultures can co-exist, but all within a broad agreement of shared norms and values that provide some guidance and direction. Larger organizations with multiple professional domains and administrative units each with their own subcultures are likely to be differentiated organizational cultures. Martin (1992) stresses that all three perspectives can co-exist in any given organization at any point in time, contingent on issues they are facing and the dynamics of power among key stakeholders.

Much research in knowledge management and organizational culture focuses on challenges faced by established organizations of considerable size and complexity (e.g., Choo, 2006; Hayes, 2018) while overlooking the particular challenges of small to medium sized organizations (McAdam & Reid, 2001). This chapter focuses on smaller organizations that are usually limited by resource constraints and working within an emergent and informal organizational culture (Durst & Edvarsson, 2012).

As with tacit and explicit knowledge creation and dissemination (Nonaka & Takeuchi, 1995), building and retaining cultural knowledge requires time and resources that smaller organizations may find lacking (Dostika & Patrick, 2013; Jarillo, 1989). Smaller organizations with limited resources may find that immediate priorities drain all available attention and energy, at the expense of strategic planning and investment in the organization's foundational knowledge (Bridge et al, 2003). As a result, time, budget and technological constraints may lead to fewer formal knowledge management channels and more reliance on informal information sharing among key stakeholders in the organization (Wong & Aspinwall, 2004).

With respect to Martin's framework of integrated, differentiated and fragmented organizational cultures (1992), it is often assumed that the more informal nature of smaller organizations coupled with the resource constraints noted above lead to fragmented cultures. This can be a strategic advantage, as smaller organizations with flatter hierarchies and more flexible processes can be more adaptable to change and be able to engage innovation unencumbered by bureaucratic procedure or organizational inertia (Cantu et al., 2009; Hudson et al, 2001; Wong & Aspinwall, 2004). However, in spending little time and energy on formal policy, procedure or strategic planning for knowledge capture and dissemination, it may be complicated for such organizations to sustain their efforts over time or to effectively leverage lessons learned from past work (Daft, 2007).

It should not be assumed however that all smaller organizations are necessarily fragmented. In smaller organizations, individuals have outsized impact on organizational culture by virtue of their larger relative influence. In some organizations, organizational culture and decision making is centered on a singular leader (Culkin & Smith, 2000). Leader-centric organizations can be highly integrated, benefitting from a high degree of consensus, clarity and consistency leading to decisions that can be made rapidly. However, should a particularly charismatic and influential leader leave, their absence may cause the organization to suddenly crash into fragmentation, with the potential for significant information loss as well. Similar situations can be found in family-owned organizations, where leadership is tied to kinship or other close interpersonal ties. Organizations whose leaders are tightly coupled due to kinship, faith, or relationship status may be cohesive but also risk being insular, risk averse, reluctant to change and prone to destabilization should emotional and interpersonal conflicts arise (Laforet, 2016; Vallejo, 2011).

In differentiated organizational cultures, countervailing checks and balances among subcultural groups can check individual power and influence, can give space and voice for other perspectives, and can strategically prepare for organizational renewal should any given leader move on. However, such organizations are also more complex, require more time to balance competing perspectives and interests,

and require significant human and financial resources to maintain. Small and medium sized organizations struggling to secure resources may find a fragmented or integrated organizational culture a more achievable goal.

KNOWLEDGE MANAGEMENT IN EMERGENT AMATEUR ORGANIZATIONS: OBSERVATIONS FROM FORMULA SAE TEAMS

The focus of this chapter is the study of engineering student project teams, particularly those building small racecars entered in annual competitions sponsored by the Society for Automotive Engineers (SAE, 2021a) and related international professional associations. Over 500 collegiate teams design, manufacture, and build a formula-style racecar based on SAE rules (SAE, 2021b), and compete annually against other teams in approximately 12 regional competitions worldwide.

Formula SAE teams face a variety of challenges in developing a stable organizational culture and foundation of tacit, explicit and cultural knowledge. Teams range in size from 6 (the minimum required to field a car in all dynamic events) to approximately 30-40 students, with most majoring in mechanical and electrical engineering programs. While it is possible to complete the project with a small core team, the annual design lifecycle requires a considerable investment of time, money and effort. Larger teams are better suited to handle the myriad of tasks and deadlines required for successful competition, at the expense of being more complex to manage. Larger teams often organize into functional sub-teams around specific technical systems, and many also include non-engineering students to cover administrative, public relations and fundraising roles. In doing so, such teams are evolving from a group driven by a small group of core members into a more differentiated and complex organizational structure.

The largest competition, held annually in Michigan, USA in May, has been running for 40 years – a considerable amount of time for teams to develop a stable organizational history, culture and knowledge base. While the competition is still open to schools fielding a team for the first time, these teams often find themselves at a considerable disadvantage, requiring a few years of experience to ramp up to a competitive level.

FSAE teams share two constraints that make them unique as small emergent knowing organizations. While small and medium sized enterprises are often considered low-turnover environments (Durst & Edvardsson, 2012), FSAE teams differ in one critical aspect. Competition rules require all participants to be active students (SAE, 2021b), and students graduate and move on to new professional pursuits. Moreover, those leaving are often the most experienced team members, having risen through the organization to hold informal and formal leadership roles. In many respects this turnover resembles those faced by collegiate sports teams and requires the same attention to succession planning and new recruit training to maintain organizational cohesion. However, as a knowledge-based team, high turnover in the FSAE context also requires attention to tacit and explicit knowledge transfer (Nonaka & Takeuchi, 1995), as well as attention to maintaining the team's cultural knowledge and history. Without such attention, turnover can create rather seismic shifts in culture – a cohesive team can quickly become fragmented if culture is tightly coupled to core team leaders, all of whom graduate at the same time. As we will see in this chapter, teams employ a variety of strategies to address this issue, with varying levels of success.

A second challenge facing FSAE teams also stems from their student status. In differentiated organizational cultures, subcultures are formed around professional specializations which hold their own norms and values. Knorr Cetina (1999) notes that domain-specific epistemic cultures are powerful forces in

knowledge-based domains, structuring how individuals engage in sensemaking, knowledge creation and decision making based on professional training. In a larger organization, it is likely for a member of a highly specialized field to have more in common with similarly trained professionals at other organizations due to their common history and continued membership in a professional community of practice (Wenger, 1998). In student teams, epistemic cultures are, at best, under development. Indeed, SAE's involvement with this event is intended to acculturate young engineering students into professional practice. At the time of their participation, however, student team members are only learning what it means to think and act as an engineer.

The same can be said for team leadership and management. Larger FSAE teams require the coordination of subteams designing interconnected subsystems, all to hard deadlines and established rules and safety protocols, and within a larger institutional culture of the university which has its own standards of practice. Handling such a complex web of challenges becomes a systems engineering question, a domain of professional practice with its own history, culture, norms and procedures (INCOSE, 2021). For many student team leaders, this may be the first time they have managed such a complex system, and it is quite feasible some might make decisions that more seasoned professionals would likely avoid.

The amateur and emergent nature of Formula SAE teams leads to fluid organizational cultures and requires a variety of approaches to handle organizational tacit, explicit and cultural knowledge. This chapter will focus on observations from 43 surveys and onsite interviews conducted by the author in the main American competitions in 2014-15 (Jones, 2017), with supplemental observations from later competition visits and contact with participants.

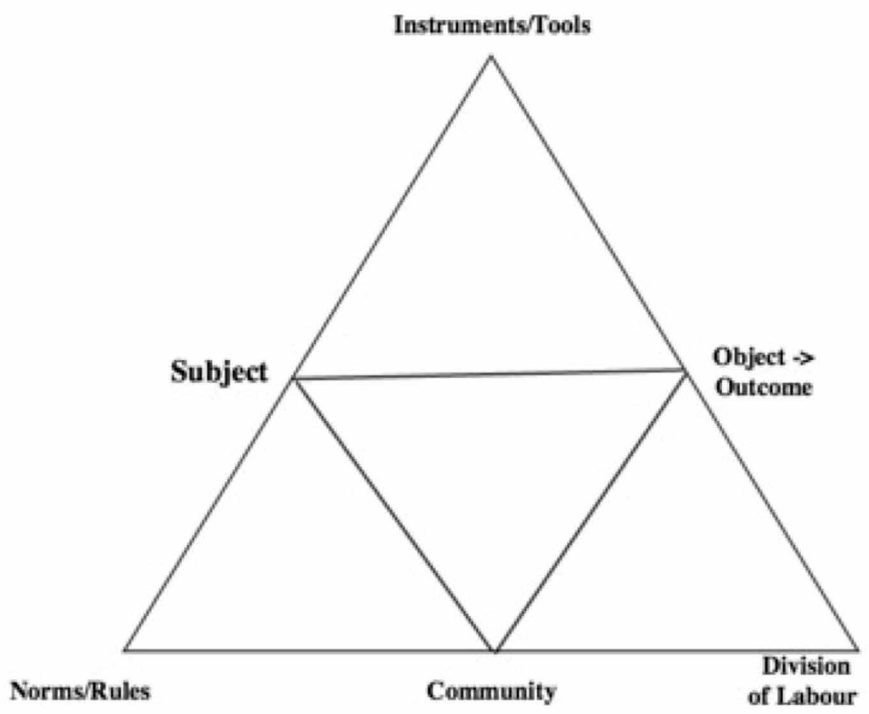
CONSIDERING CONTRADICTIONS: A CULTURAL HISTORICAL ACTIVITY THEORY APPROACH

Research questions were informed by cultural-historical activity theory (CHAT) (Engestrom, 1999; Engestrom, 2008; Leont'ev, 1978). The central focus of CHAT is a core activity where human subjects use various tools and technologies to create objects that ideally meet their intended outcome (Vygotsky, 1978). CHAT extends Vygotsky's constructivist dialectical model of activity to include broader socio-cultural influences such as unwritten norms and formal rules, a community of external stakeholders, and divisions of labor and power. CHAT descriptions of human activity are commonly represented as a diagram shown below in Figure 1 and has been widely used in a variety of information science research domains (Allen et al., 2011).

Of particular interest here is the notion of contradiction (Engestrom, 2008; Meyers, 2007). In attempting to realize any given outcome, there can arise points of conflict within and among nodes in the diagram above. Negotiating contradictions and competing priorities can be essential in collectively achieving the intended outcome, and conversely, failure to attend to specific contradictions can be detrimental to the overall outcome.

There are four types of contradiction noted in CHAT (Engestrom, 2008). Primary contradictions focus on conflicts within any given node (e.g., in a group setting, two subjects may approach the activity differently, and have to reconcile their differences to collaborate), whereas secondary contradictions investigate conflicts between various node connections (e.g., a subject may choose three different research methods to investigate a given problem, yielding three related but distinct results that then have to be reconciled).

Figure 1. A generalized cultural historical activity theory (CHAT) diagram (adapted from Engestrom, 1999)



There are also contradictions that occur between various activities. Tertiary contradictions occur when a given activity changes over time (e.g., an initial plan for an activity may have to change as an organization gains or loses resources or contends with changes in external rules and regulations). Quaternary contradictions occur when one activity model shares an intended outcome with another activity model (e.g., if two groups seek the same outcome, it may not be possible for both to be successful, setting up potential for conflict).

The concept of contradiction is powerful in framing human activity because it does not presuppose any ideal or correct answer. There are often multiple potential paths through the activity model, and while some may prove more fruitful than others, the CHAT model allows for a range of interpretations and responses to the contradictions that arise.

For the purposes of this chapter, we will examine three particular contradictions – organizational turnover (a tertiary contradiction between a team's current and future roster), capturing and retaining organizational knowledge (mostly involving secondary contradictions between subjects and various information systems), and complex multi-year and collaborative design lifecycles (for multi-year projects, a tertiary contradiction in extending the normal design cycle over time; for collaborative projects, negotiating quaternary contradictions as competing teams collaborate to meet the same goal). Unless otherwise noted, quotations below are derived from and represented by respondent number, a full table of which can be accessed in (Jones, 2017).

Organizational Regeneration in a High Turnover Context

Given that FSAE teams are made up of student team members and leaders, it is predictable that a number of students will depart the team annually due to graduation. It is common for teams to report annual attrition rates of 50% or more. Such attrition can be particularly stark at the level of student team leadership, most of whom are senior students. While teams are well advised to attend to this inevitable tertiary contradiction, it is surprising how many treat this as an afterthought. When asked about how their team plans for organizational regeneration, one faculty advisor honestly noted "it just happens" (#40).

Such an attitude leaves quite a lot to chance and does not seem to be successful – through analysis of competition results (SAE, 2021c) #40's team routinely finishes in the lower half of the competitive field of 120 in Michigan, and has never finished in the top 25 despite being a regular annual participant. While their team's culture might be satisfied with such mediocre results, other teams may prefer a stronger return on what is a significant investment of time and effort.

Other teams are more deliberate about team recruitment and regeneration. Respondent #29 noted that their team identified specific roles that would be left vacant due to graduation and worked to recruit specific people to fill those roles. Respondent #38 took formal recruiting a step further, noting a points system based on performance goals that helped determine final team rosters in any given year.

Six respondents noted that they recruited team members just before the competition to give new team members a view of the final intended objective of the team. Those members may not have a large seniority advantage over those recruited in the new academic year, but the experience and context they gain from witnessing the competitive environment can be both elucidating and invigorating and provide context for their actions.

Structured recruiting is not the only approach used by FSAE teams. A similar number of teams preferred a more organic approach to team recruitment. Respondent #24 noted:

We allow them to make the call – if they're interested in doing things, we'll see more of them. If they just signed up for a club because they thought it was cool, we'll see less and less of them as they realize real work is expected. By the end of the year, we know who's on and who's not. (#24)

Such an emergent model was also common in determining team leadership – instead of ascribing roles and titles in advance, respondent #4 noted "we acknowledge individuals that put in more effort. They in turn get consulted more and become obvious unspoken leaders." In such a model, new team members rise through the ranks through dedication to the team's overall mission. Such a model may, however, recreate a homogeneous team culture where only those who "fit" the dominant organizational culture are appreciated, potentially marginalizing students and skill sets that could be valuable in diversifying the organization.

With respect to development of new recruits, 10 teams noted the importance of mentoring, at various levels of formalized implementation. Mentorship and apprenticeship leverages the socialization aspect of Nonaka and Takeuchi's socialization, externalization, combination and internalization (SECI) model of knowledge creation (1995) by creating semi-structured opportunities for students to learn from their peers through social contact. Respondent #7 notes:

A lot of my time this year will be sitting with the new recruits to quickly bring them up to speed with the team and general suspension design. I have also given all the new members senior mentors. Hopefully

this will allow for a better knowledge transfer to the freshmen. We are also looking at ways where the new members can sit, watch and ask questions while the senior members design parts, so they can see more complicated design process. My hope from this is that this will show the new members how to make good design decisions, and get a general idea of how more complicated parts are designed. Technical knowledge can always be learned, but this design mentality can only be gained from experience. (#7)

Such an approach renews membership in important sub-systems that are essential to a successful car. Structuring early exposure to subteams allows senior members to mentor new members in the context of practice of their activity, thus sharing tacit knowledge gained on the job. This can be seen as an example of Lave and Wegner's (1991) notion of legitimate peripheral participation, in which peripheral activity can be seen as a valuable learning channel for less skilled members.

As with most CHAT contradictions, there is no one "right" path for recruiting and developing team members, and processes usually derive from their fit with the overall organizational culture. A more structured recruitment strategy may better fit teams with a more developed, complex and differentiated organizational culture, whereas teams operating in a more fragmented and emergent organizational culture may opt for lower-overhead solutions at the risk of having significant skills and organizational knowledge depart with little to no strategic plan for regenerating an effective team in future years.

Capturing and Retaining Organizational Knowledge

While sharing tacit information through mentorship is effective, it is also important for teams to be able to record past developments to learn from their successes and mistakes. Creating and maintaining a permanent record of knowledge allows teams to move up the SECI knowledge creation model from socialization to externalization and combination (Nonaka & Takeuchi, 1995) and can provide a rich archive of internal information to inform future decision making.

Not surprisingly, team reports, image collections and data from internal testing were commonly cited by team members as resources that inform design decisions. However, many teams noted difficulties in using their internal information repository. Respondent #17 noted the following about their team's image collection:

We have a collection of images on our server from competition photos – but they're a bit all over the place. A photo might not say much – a photo of a brake system doesn't tell you which brake system it might be, and no one seems to write down notes about that, especially competition photos. It's easy to forget whose car was photographed – we try to tell our photo teams to take photos in an order with the car # and inspection stickers so we can at least trace them to specific cars/years, but that's only followed sometimes. (#17)

Similarly, respondent #29 noted that internal team reports were of questionable accuracy and quality, and that it was often easier to track down the original author to ask questions. Another noted internal testing data as "...garbage – half the data was missing, the rest of it was just crap. I hope I can do a better job this year, definitely going to be a summer project." (#14)

This speaks to an essential functional role often acknowledged but not always supported by team organizational culture. Creating quality data is not automatic. Folders filled with untagged digital photographs or poorly formatted data from testing quickly become detached from context, making their

relevance for future knowledge creation and decision making questionable. Especially since student engineers may have limited experience with technical writing, documentation or archive management, leaving students to their own devices without training or support may create a range of information artifacts of questionable future value. Without team members committed to the curation and management of the team's information archive, a virtual shoebox of disparate and unlabeled data sources may prove to be a source of confusion andinformation overload, making its use more complicated in future decision making.

Also worthy of note from #14's comment is the issue of time. Poor information repositories are often the consequence of inadequate time to faithfully document events. Especially in more fragmented organizational cultures, during the design lifecycle team members are frequently bouncing from task to task, many of which are of immediate and critical importance. Documentation is considered as something that can be done later, a time that conveniently never arrives. This is especially the case with data collected at competition, which can be a chaotic and exhausting time. Without time reserved for combination and internalization, information loss is very likely and that can directly impact the quality of the team's information repository especially as the original authors graduate and move on to other pursuits.

All that noted, there are also success stories. Respondent #3 noted that their team ran a cloud-based information repository that allowed team members to contribute annotations about past team reports and data, both in their lab facilities and on-site at competition. Intriguingly, this information repository also served as a connection point with team alumni, who had continued access after graduation and often contributed observations. Unlike other organizations where departing team members may be reluctant or contractually unable to share their expertise, recent graduates are more willing to be of assistance to the generations of team members that replace them. Indeed, seventeen respondents noted alumni connections as a source of information in their research and development work – and a team information repository that has regular presence and participation from recent alumni can be a dynamic and valuable team asset.

While there are significant challenges in creating an effective and robust information record of team data in an amateur and high turnover context, the benefits are also clear. Arguably the gold standard in recent years is Global Formula Racing (GFR) – a complex, differentiated international team run jointly by Oregon State and DHBW Ravensburg in Germany (GFR, 2021). Team members design subsystems collaboratively from both locations, building separate cars on two continents from a collective information database. While the author's initial interview with the team's faculty advisor was cut short due to the team's scramble to resolve a technical issue causing them to fail the required competition noise test, it was clear from observation the GFR team was adept a integrating just-in-time access to information, with team members carrying tablets to access their repository in real time, with the faculty advisor monitoring progress and ensuring that important information about the noise issue was added to the organizational record for future analysis.

Such dedication to recording and managing data would be moot if it did not yield results – however, GFR has become a dynastic team in both American and European competition venues, winning the Michigan competition 5 times in the 2010s and finishing outside of the top 5 only twice due to technical failure in the main endurance race (SAE, 2021c). This record of success has attracted the attention of competing teams, with five respondents specifically noting they were striving to imitate GFR's comprehensive information systems foundation. However, many also noted technical, financial and human resource challenges in doing so. Respondent #21 noted some challenges faced being an international team attempting to operate on American soil.

We're a lot more tied to the cloud than we realized – it's been frustrating to not have 24/7 access to data. One team member took it all offline yesterday at the hotel so we at least have everything on a laptop in the truck, but that's not easily retrieved everywhere. And while a few of us bought a data plan before getting here we're already over - a few of us are going have a shock on our bills when we get back. (#21)

While an information repository requires access to computing and wireless networks, arguably more important is an organizational culture that supports and privileges its continual use in information seeking, knowledge creation and decision making. As teams mature and develop consistent organizational structures and cultures, they are more able to emulate the impressive infrastructure put in place by top-tier teams like GFR. Newer teams are likely to find such a project a considerable challenge to take on, as are teams that cannot secure the human, financial and technical resources required to develop and refine such information systems.

Supporting Multi-Year and Collaborative Projects

Information repositories can assist with another common challenge faced by FSAE teams. The design cycle of an FSAE car is rapid, moving from initial design to final testing and validation at competition in one academic year. While systems can undergo iterative improvement over time, more profound changes (e.g., changing from space-frame chassis design to integrated composite tubs, or integrating full aerodynamic packages) may not be feasible in this compressed design cycle.

To successfully manage multi-year projects, student leadership must juggle the demands of a longer than usual research and development phase, potential conflicts with other critical subsystems of the car, high turnover on team rosters, potential annual changes in competition rules, and the often tough decisions of telling some team members that their efforts will not make a given year's car because the system is not yet fully developed. This creates a number of potential tertiary contradictions as the activity evolves of time and circumstances. Building a robust information repository of past design and test data as well as access to the tacit knowledge of past members becomes even more important in managing long-term complex systems change.

Given these challenges, it was not surprising that seven respondents suggested their team avoids the problem altogether by only focusing on projects that can be completed within a given year. While easier, this may mean certain innovations are off-limits or that systems may be rushed to meet immediate deadlines, a strategy which may increase odds of systems failure during the critical endurance race. Three respondents simply said they just work extra hard – but such a solution only works if there's extra time and energy to spare. Such an approach is likely to be more common in teams with a fragmented organizational culture coping with the crises of the day and unable to scale up to challenges that do not match their timeframe.

A differentiated organizational culture with a strong knowledge base is more likely to be able to adapt to the significant logistical challenges of multi-year projects. Respondent #4 noted a more sophisticated multi-year process.

What approximately 15 1-3rd year members will do is work on multi-year development projects. This allows for implementation of larger changes to take place in more realistic time frames. The individuals assigned to these areas will become extremely familiar with the new systems they are designing, and then transition roles to implementation when their design is being implemented on the car. (#4)

By starting newer younger team members out during early stages of design, they are able to follow the design cycle throughout their tenure and still gain the satisfaction of seeing their work fully implemented. One potential drawback of this strategy, however, is that younger team members may not be sufficiently knowledgeable to lead a complex design effort. Early curriculum in engineering programs covers foundational concepts, with specialized domains only covered in later years of the program. Leaving sophisticated design projects to the inexperienced may lead to the creation and implementation of partial, incomplete or wrong mental models, a common problem in weakly scaffolded project-based learning (Mandin et al., 1997). An apprenticeship model lead by more accomplished students might be a more fruitful approach than assigning new team members complex design projects without scaffolding and pre-requisite knowledge.

A more drastic solution can be skipping a competitive year and fielding a car that has been in design and development for two or more years. Sometimes this happens by necessity when a team's annual design cycle leads to an underdeveloped car that the team agrees is not ready for competition. Planning ahead for a gap year may be less disappointing and a better strategy for the organization as a whole. Indeed, new teams often prepare for two or more years, visiting competition as observers and engaging in discussion with more established teams before committing to their first design cycle.

Such a strategy works because of a rather peculiar cultural convention of this competitive context. Competition rules structure a zero-sum game – points in all events are gained in ratio to other teams' results (SAE, 2021b). It would be fair to think such a point structure would generate a quaternary contradiction and that hoarding of information and suspicion towards competing teams would be the norm.

However, while the rules of the competition may discourage sharing, the norms of the overall circuit are cooperative with many instances of information sharing among teams evident. Respondent #10 sums it up thusly:

During competition, we've helped out teams by lending them or giving them tools or materials. It's in the spirit of the competition to help each other out, that's part of the beauty of FSAE. We helped [X] University modify their silencer which allowed them to pass sound tech. We've given several teams our endurance event data and given or traded other items with dozens of other teams. (#10)

This is not an isolated case – 17 teams in the sample shared examples of direct assistance to other teams at competition or sharing data and design with teams are in direct competition for the same end goal. Assistance to new teams or those considering joining the circuit is particularly common and makes sense given the emergent teams are unlikely to be direct threats, at least in the short term. Established teams are often happy to assist newer teams get up to speed.

That noted, there is a certain etiquette to information sharing. Respondent #23 noted significant frustration with one new team who swarmed their competition paddock taking detailed photographs of their car, noting:

What's weird is that if they just asked us questions, we'd probably give them better information. Not sure what a bunch of photos without the background design work will get you. One team I swear might come up with a close copy of our car next year. They won't know why, and it probably will never work, but it just might look exactly the same. (#23)

Beyond ad hoc assistance, teams also occasionally collaborate in more formal arrangements. The aforementioned GFR team and University of Stuttgart (Pistorius, 2014) have both created structured collaborative efforts that increase their overall capacity. These collaborative efforts have paid off – GFR and Stuttgart have become the dominant teams at the Michigan competition, having won all competitions in the 2010s (SAE, 2021c). Respondent #1 noted their already quite successful team was considering a "sister school" arrangement with a similarly competitive team to emulate the track record of these teams. It remains to be seen if other schools develop similar joint ventures.

As small organizations with limited technical and financial resources, it also may make sense for teams to collaborate on particular technical challenges. A long-standing example of this would be the Tire Testing Consortium, a collaborative research effort funded by small contributions from over 650 emergent and established teams worldwide (MRA, 2021). Tires are the foundation of any suspension design model, but tire data from manufacturers is sparse due to limited production runs of small car tires and industry secrecy. While teams can conduct their own tire data acquisition, this requires specialized equipment most schools do not possess and a significant budget, as it requires wearing tires down to failure. The consortium was created by teams in collaboration with a major suspension design judge to fill this gap – the cooperative hires a professional testing facility that generates tire test results shared by all member teams, allowing member teams to make better data-driven decisions on suspension design. Such co-optition (Gnyawali & Park, 2009) enables small organizations to gain access to valuable resources that they alone would not be able to acquire, and is one of many examples of cooperation and collaboration evident in the FSAE competitive context.

SOLUTIONS AND RECOMMENDATIONS

FSAE student teams face a myriad of challenges as they negotiate the contradictions of their chosen activity. However, they do not do so in a vacuum – they exist as part of a larger educational institution that can provide significant space, technical and financial resources. While teams with a strong foundation of support from their schools are not automatically successful, they are likely to face fewer challenges than less resourced teams who are forced to spend more of their limited time securing basic resources or battling against a contrarian administration. Attaining such support, however, is not guaranteed.

A potentially important role in negotiating this contradiction is that of faculty advisor. Without support from senior leaders, knowledge management efforts are often ineffectual (Davenport et al, 1998). In this particular context, however, student leadership is temporary and fluid due to graduation, and may find themselves overwhelmed with the immediate task at hand. A committed faculty advisor can help by developing the team's longer-term organizational priorities irrespective of generational shifts in leadership, which can include sustaining the team's tacit, explicit and cultural knowledge and mentoring student team leaders to prioritize knowledge creation and retention. In doing so, an engaged faculty advisor can act as a de facto chief knowledge officer (Liebowitz, 1999), playing a role that student leaders may struggle to fulfill.

While all teams nominally have a faculty advisor, their level of involvement varies considerably. Indeed, many are not present at the competition even though rules strongly suggest they must be (SAE, 2021b). Respondent #28 even struggled to name their advisor, noting that they played more an administrative role but otherwise let the team "do its thing." Their results over years of competition suggest that such a

laissez-faire approach was not particularly helpful. Indeed, seven teams noted some frustration with the level of engagement with their current advisors, wishing for more support and mentorship.

Less common but equally problematic are faculty advisors who play too active of a role. Respondent #22 suspected that a competing team had a faculty advisor who did all the design work, relegating team members to more semi-skilled roles. This allegation was later substantiated by informal discussion with a competition judge and bears out in competition results. The alleged team's performance in the design event, where students must answer questions directly and advisor participation is explicitly banned, has significantly trailed their overall results. When forced to defend their design, team leaders and members did not have a solid command of their own work and that judges were quite aware of the disconnect.

Advisors that seek a more balanced and measured role do exist and are generally more appreciated. Respondent #2 noted their advisor plays a largely hands-off role, but is critical in acting as a surrogate for design review judges, asking tough questions in advance and preparing sub-team leaders well for extensive questioning of their work. This insight was not surprising as Respondent #2's team is routinely strong in the design review component of the competition.

Respondent #3 said their advisor allows team leaders to forge their own direction, interfering only when decisions may run afoul of university policy. He also noted their faculty advisor was himself a former FSAE team leader, and served as an excellent mentor for both engineering, leadership and general life skill development as a result. Similarly, respondent #38, a faculty advisor, noted their approach as simply "I would only intervene if safety or budget were at risk. They are empowered to fail."

Achieving the optimal balance between micromanagement and absentee status is a difficult balancing act. Golding (2009) notes a continuum of constructivist learning between unstructured and teacher-driven approaches, privileging a middle ground community of inquiry model where faculty allow for significant student direction and leadership but within a framework of disciplined dialogue, mentorship and critique. Golding also notes that faculty habituated to a more authoritative role might struggle with this ideal state - however, within the research record at hand here, it was clear respondents suffered more from a lack of engagement than over-engagement.

Given the many professional responsibilities of faculty members, it is understandable how many might prefer a more limited role. Being an effective facilitator of project-based learning is time consuming and often not well supported (Rosenfeld & Rosenfeld, 2006). However, in a high-turnover environment, faculty advisors can play a critical role in bridging multiple generations of team rosters and reflecting the team's overall history and culture. An advisor who strikes the right balance between "empowering to fail" and necessary intervention allows student team members and leaders to engage complex knowledge generation activities while protecting the team from situations that cause them to run afoul of larger institutional concerns.

Faculty advisors can also be effective bridges to the larger institution and help secure the technology, space, budget and technical assistance required to better facilitate the activity of their team. Respondent #37, a faculty advisor, noted:

For space, the team has space within what is called the [X], a 24,000 ft^2 space dedicated to the support of our student engineering competition teams. Each team within this space (there are currently 7 hosted) has office space, dedicated build and storage space, and access to the common machine tools. X is open 24/7. It has machine tooling available 24/7, up through 3-axis CNC mill. The school staffs a ½ time machinist/staff support person dedicated to the infrastructure support of the X and its teams. (#37)

This is arguably a gold standard – most teams do not claim access to such a large custom-built facility available 24/7 and monitored by technical staff, and not surprisingly, #37's team is a perennial top-tier contender in the Michigan competition. However, such facilities do require negotiating for a large share of a school's budget, which is a task that might be daunting for student team leaders. A strong faculty advisor can play a valuable role in negotiating for such support with key institutional stakeholders.

There are reasons why schools will provide such support. FSAE teams often play key roles in recruiting new students to the school, securing alumni fundraising, and meeting the work-integrated learning and soft skill training requirements of accreditation agencies (e.g., ABET, 2021). Teams are well advised to maintain mutually beneficial relationships with their administration and try their best to negotiate conflicts with a bureaucracy that is often at odds with their emergent, amateur and at times risky organizational culture. A strong faculty advisor can be a good guide, mentoring team leaders on how to best navigate complex institutional procedures and smoothing over inevitable tensions.

However, resource issues are a common concern, especially when compared to other collegiate competitive domains. Respondent #4 notes:

It is in our opinion that the design teams are like the engineering varsity sports teams, though the actual varsity sports teams are catered to significantly more… I spoke with teams from Germany as an example, where they do not have a varsity football team, so they have everyone join and get excited about their team. (#4)

While FSAE teams are similar to collegiate varsity sports teams in structure, collegiate sports teams are often considerably better supported with special-purpose facilities and dedicated coaching, especially in the American context where varsity sports acts as a de facto minor league for professional sports and are a considerable revenue generator for the university. While in the short term it is unlikely that schools would support competitive engineering project-based learning teams to the extent they support their basketball or football programs, schools that recognize the value of their project-based learning teams and support them in the form of faculty advisors and infrastructural resources are providing their teams with a context more conducive to success. However, achieving this requires both significant commitments from the school, faculty and staff as well as stronger integration of team culture and practice into overall school priorities.

FUTURE RESEARCH DIRECTIONS

The dynamic nature of FSAE as a competitive environment assures that this chapter and the larger work that serves as its foundation are partial and subject to change. Two considerable changes are impacting the FSAE competitive environment at present. As gasoline-powered internal combustion engines are arguably reaching the end of their lifecycle, many teams are switching over to electric drive, which is a considerable systems engineering challenge and requires a teams with significantly different skill sets. This switch is unlikely to be done in a single annual design cycle. Investigating how established teams are handling (or avoiding) this change would be an excellent opportunity for future research. It is feasible to suggest newer teams with more fragmented cultures and less of a vested knowledge base could leapfrog more established teams with substantial – but increasingly obsolete – knowledge.

The second change is one we all have had to contend with – the global COVID-19 pandemic. Both the 2020 and 2021 North American competitions were forced into virtual space to comply with local health and safety requirements, and by most accounts this replacement is not a compelling alternative. While some courses and research projects can adapt to virtual delivery reasonably well, FSAE is by design and history an experiential activity. Participants are literally getting their hands dirty in creating a fully operational racecar, and the majority of competition points are tied to dynamic events that test an actual car's on-road performance. This cannot be simulated without drastically altering the nature of the core activity. Sadly, two years of suspended competition have robbed contemporary student team members of a richer professional development and educational experience.

How the series bounces back after COVID-19 restrictions are lifted would be an interesting question for future research. It can be assumed that teams with a strong organizational culture and internal knowledge base can reset to their 2019 effort with relative ease. It is also likely that teams with a culture oriented to strategic planning may take advantage of this hiatus by committing to complex, longer-term projects while free from the many immediate pressures of fielding a competition car. Conversely, teams with a more fragmented organizational culture may struggle to maintain cohesion without an immediate project at play, and if significant changes in enrollment influences overall school budgets, other teams might find themselves particularly starved for resources.

While this research is focused on teams in the Formula SAE series, observations from this series would extend well to similar project-based learning opportunities, including others supported by the SAE (SAE, 2021a) as well as similar competitive domains in other academic disciplines. This research is also instructive in the use of CHAT as a way of framing complex organizational development challenges beyond the domain of ad-hoc teams and capstone design courses which are the primary focus of much project-based learning research. Beyond the educational domain, this research may also transfer well to other smaller teams with high organizational turnover, such as retail or hospitality teams or entrepreneurial startups operating on limited resources and weaker cultural knowledge foundations.

CONCLUSION

Formula SAE teams are small emergent knowing organizations engaged in an intense design cycle shaped by significant information seeking, knowledge creation and decision making challenges. These challenges are more complex when one takes into consideration high organizational turnover due to graduation and the amateur status of students as both engineers and leaders of the organization. Building a strong differentiated organizational culture with heterogeneous and complementary roles and a strong commitment to knowledge management helps established teams learn from past efforts and develop plans for future R&D, including more complex design challenges that might require sustained effort or collaboration with other partners. While student team members and leaders handle these challenges well, teams that operate within a supportive institutional environment and under the guidance of effective faculty advisors face fewer challenges in their quest to design, manufacture, test and field a quality competition-ready car.

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KEY TERMS AND DEFINITIONS

Community: An extension of the core activity in CHAT, community refers to external stakeholders who are not subjects of an activity but nevertheless have a say in its realization.

Division of Labor: An extension of the core activity in CHAT, division of labor represents relations of power and authority that may enable or constrain the execution of a given activity.

Norms/Rules: An extension of the core activity in CHAT, norms/rules represent both formal laws and informal customs at play as subjects consider their activity.

Object/Outcome: Part of the core activity in CHAT, the object is the end result from subjects engaging tools/technologies to complete an activity; the outcome is the intended object, which may differ.

Primary Contradiction: Conflicts that may arise within one node of the CHAT model (e.g., inconsistencies among formal rules and unwritten norms/values that may require reconciliation).

Quaternary Contradiction: Conflicts that arise when one activity model clashes with another that share similar intended outcomes (e.g., two groups independently working towards a goal that only one group can attain).

Secondary Contradiction: Conflicts that may arise between various nodes of the CHAT model (e.g., a subject's intended activity raising concern of community members).

Subject: Part of the core activity in CHAT, the subject is an individual or group attempting to achieve a given activity.

Tertiary Contradiction: Conflicts that unfold as a CHAT modeled activity changes over time (e.g., changes in subjects, rules, tools/technologies over time may force a change in the overall execution of the activity).

Tools/Technologies: Part of the core activity in CHAT, tools/technologies are methods engaged by subjects to realize their activity.